Week 13 Assignment

Mixed-effects models II

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf`, along with your Data Safari submission, via Canvas before Week 14's lecture (Monday). See the syllabus for the late policy.

Note: This week also includes your final Data Safari exercise — a separate, short assignment. Both are due at the same time.

Part 1: Random slopes

Using this week's tutorial's `field_data` and models:

1. Compare `m_mixed` (random intercept only) and `m_slopes` (random slope) with `anova()`.

Does the random-slopes model fit significantly better?

2. In 2–3 sentences: what would it mean, biologically, if the treatment effect really did vary meaningfully from site to site?

Part 2: Crossed vs. nested

For each scenario, write the `lmer()` random-effects structure you'd use, and explain whether it's crossed, nested, or a mix of both:

1. Blood samples taken from 4 individual animals, each sampled at 3 time points.
2. Soil cores taken from 5 plots nested within 3 sites, with plots physically distinct per site.
3. Behavioral observations made by one of 3 observers, where each observer records data at every one of 6 field sites.